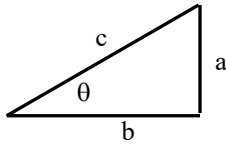


This is a closed book and closed notes group problem with no connection to any source of information. Calculators are permitted only to do mathematical calculations. No devices that can contact the internet are allowed. The **ONLY** physics formulas that may be used are those given below. Define all symbols and justify all mathematical expressions used. Make sure to state all assumptions & approximations used to solve a problem. If this were a quiz, partial credit would be given for a well-communicated problem-solving framework based on correct physics. Each problem is worth 25 points: **In the context of a unified solution**, credit will be awarded as follows: a useful sketch that shows all relevant quantities, listing the known quantities, and writing the quantity to be determined is 7 points; writing the physics principles useful to solve the problem, writing any assumptions or approximations needed, drawing useful physics diagrams, and writing the equations that apply the physics principles in terms of the quantities from your sketch is 6 points; planning the solution by describing how to use those physics equations to construct an algebraic answer is 5 points; solving those equations for a symbolic result is 3 points; checking the units & reasonableness of the symbolic result is 2 points; calculating a numerical result & stating why its units are correct and giving a reason why that result is not unreasonable is 2 points. If no numerical answer is required, 1 point will be added the planning the solution and 1 point to solving the equations and checking the units & reasonableness of the solution.

Useful Mathematical Relationships:



For a right triangle: $\sin \theta = \frac{a}{c}$, $\cos \theta = \frac{b}{c}$, $\tan \theta = \frac{a}{b}$,

$$a^2 + b^2 = c^2, \sin^2 \theta + \cos^2 \theta = 1$$

For a circle: $C = 2\pi R$, $A = \pi R^2$

For a sphere: $V = \frac{4}{3}\pi R^3$, $S = 4\pi R^2$

If $Ax^2 + Bx + C = 0$, $x = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$, $(a + b)^2 = a^2 + 2ab + b^2$

Fundamental Concepts, Principles, and Definitions:

$\sum \vec{F} = m\vec{a}$	$\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$	$a_{x \text{ average}} = \frac{\Delta v_x}{\Delta t}$	$v_{x \text{ average}} = \frac{\Delta x}{\Delta t}$	$\tau = rF_{\perp}$
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Under Certain Conditions:

$F = mg$	$F = \mu N$	$F_{\max} = \mu N$	$F = k\Delta L$	$\sum \vec{F} = 0$	$\sum \vec{\tau} = 0$
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Useful constants: 1 mile = 5280 ft, $g = 9.8 \text{ m/s}^2 = 32 \text{ ft/s}^2$

Your team is tasked with investigating shock absorbers for buildings in an earthquake zone. To test one design in the lab, you plan to support a uniform horizontal steel girder by putting it on two springs attached to a horizontal floor. To purchase the springs for the test, you need to specify how their spring constants are related to each other. Because of the constraints of the building design, one spring will be at one end of the girder and the other will be $\frac{1}{4}$ of the girder's length away from the other end. The springs must be the same length before they are compressed by the beam. You know the girder's length, weight and each spring's mass.